

$$X = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix}, y = \begin{pmatrix} 5 \\ 6 \end{pmatrix}, n = 2, d = 0.01$$

$$m = \begin{pmatrix} -1.3680 \\ 1.0956 \end{pmatrix}, b = \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$$

$$\hat{y} = X_m + b = \begin{pmatrix} 1 & 3 \\ 4 & 10 \end{pmatrix} \begin{pmatrix} -1.3680 \\ 1.0956 \end{pmatrix} + \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix}$$

$$\hat{y} = \begin{pmatrix} 1.95178 \\ 5.4800 \end{pmatrix} + \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix} = \underline{\underline{\begin{pmatrix} 2.98818 \\ 6.3787 \end{pmatrix}}}$$

Calculate error (e) and Cost (J)

$$e = \hat{y} - y = \begin{pmatrix} 2.98818 - 5 \\ 6.3787 - 6 \end{pmatrix} = \underline{\underline{\begin{pmatrix} -2.01182 \\ 0.3787 \end{pmatrix}}}$$

~~$$J = \frac{1}{2}$$~~

$$J = \frac{1}{2} \sum e_i^2 = \frac{1}{2} \left((-2.01182)^2 + 0.3787^2 \right)$$

$$= \frac{1}{2} (4.0474 + 0.1434) = \underline{\underline{2.0954}}$$

Date

$$\Delta J = X^T e = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -2.0458 \\ 0.3787 \end{pmatrix} = \begin{pmatrix} -0.5310 \\ -2.3504 \end{pmatrix}$$

Gradient ΔJ

$$\frac{\partial J}{\partial m} = X^T e = \begin{pmatrix} 1 & 4 \\ 3 & 10 \end{pmatrix} \begin{pmatrix} -2.0458 \\ 0.3787 \end{pmatrix} = \begin{pmatrix} -0.5310 \\ -2.3504 \end{pmatrix}$$

$$\frac{\partial J}{\partial m} = X^T e$$

$$\frac{\partial J}{\partial b} = e = \begin{pmatrix} -2.0458 \\ 0.3787 \end{pmatrix}$$

$$C_{\text{new}} = C_{\text{old}} - \alpha \Delta J$$

$$w_{\text{key}} = \begin{pmatrix} -1.3690 \\ 1.0956 \end{pmatrix} - 0.01 \begin{pmatrix} -0.5310 \\ -2.3504 \end{pmatrix} = \begin{pmatrix} -1.3637 \\ 1.1191 \end{pmatrix}$$

$$b_{\text{new}} = \begin{pmatrix} 1.0364 \\ 0.8987 \end{pmatrix} - 0.01 \begin{pmatrix} -2.0458 \\ 0.3787 \end{pmatrix} = \begin{pmatrix} 1.0569 \\ 0.8949 \end{pmatrix}$$